



Cooperative Multi-Agent Constrained POMDPs: Strong Duality and Structural Equivalences

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Joint work with **Nouman Khan** (Amazon) ([CDC 2023](#), [CDC 2025](#), [Ph.D. thesis](#)), Hsu Kao (JP Morgan) ([AISTATS 2022](#))

Acknowledgments

□ Nouman Khan

□ Hsu Kao

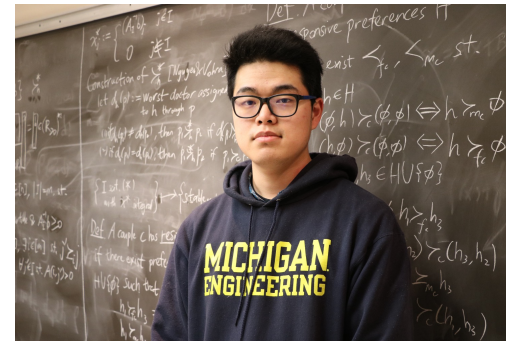
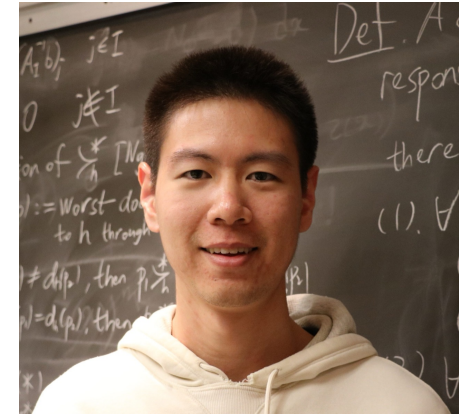
□ Dengwang Tang

□ Aditya Mahajan

□ Demos Teneketzis

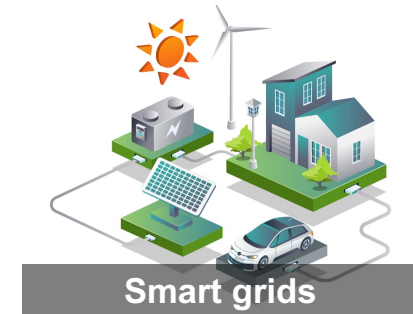
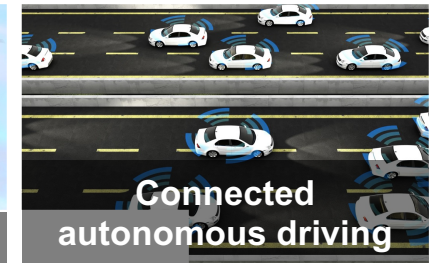
□ NSF

□ ONR



Decision-Making in Multi-Agent Systems/Networks

- ❑ **Multi-Agent Systems:**
 - ❑ A group of agents (decision-makers) who interact with each other in a shared environment
 - ❑ Present in a variety of forms and in diverse applications
- ❑ Extremes of possible configurations:
 - ❑ **Cooperative:** TEAM (Common goal) (*MarschakRadner1972*)
 - ❑ **Competitive:** GAME (Individual goals) (*FudenbergTirole1991*)
- ❑ Unique Challenges due to a combination of:
 - ❑ Multi-agent interactions
 - ❑ Complex information structures:
 - ❑ Information asymmetry and partial observability
 - ❑ Joint constraints
- ❑ Focus of this talk: Cooperative setting of a TEAM with joint constraints



Background: Sequential Decision Making and Information States

- ❑ Agents need to use information increasing over time --- (Private) History
 - ❑ Increase of complexity and memory
- ❑ *Information State-Compression of history sufficient for decisions (Witsenhausen 1976)*
 - ❑ Simplifies search for (optimal or good) control policies + yields algorithms (DP)
 - ❑ Enables learning-based solutions + algorithms (RL)
- ❑ Examples:
 - ❑ MDPs --- Current state (*Kumar Varaiya 2015*)
 - ❑ Model and Strategy Independent
 - ❑ POMDPs --- Belief state (nonlinear filter) (*Astrom 1969, Smallwood Sondik 1973*)
 - ❑ Strategy Independent but Model Dependent
 - ❑ Model and Strategy Independent *Information State* exists that increases in time
- ❑ *How about Information States in multi-agent teams?*

Challenges for cooperative multi-agent problems

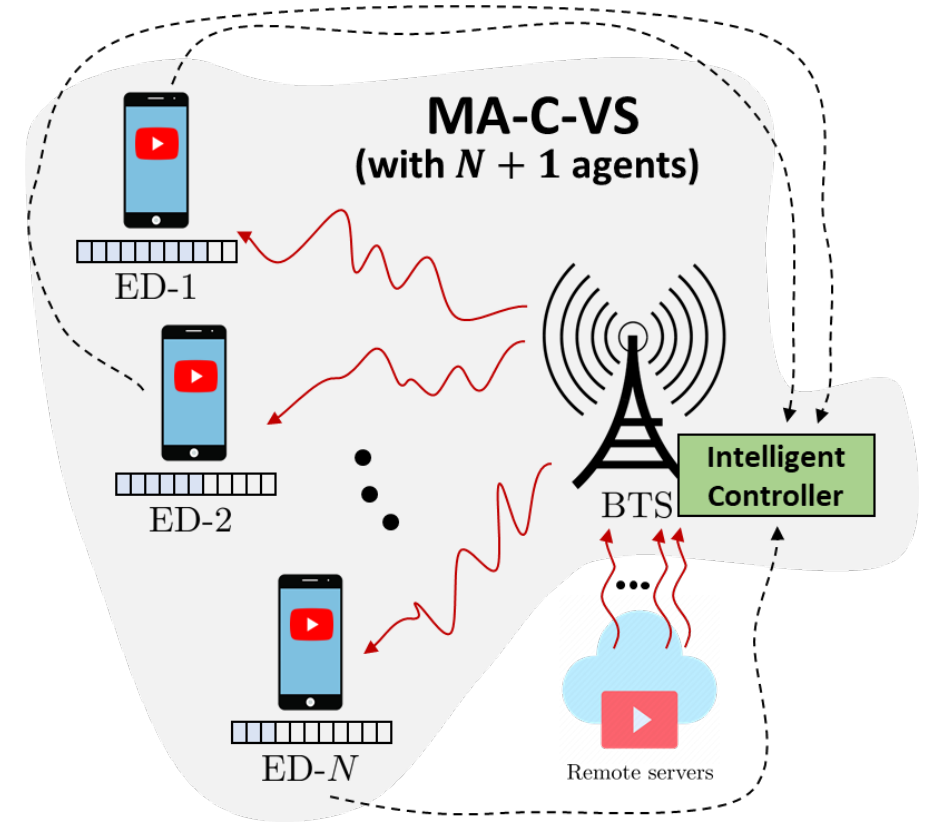
- ❑ Information asymmetry: Each agent works with their private information
 - ❑ Agents need to use complex reasoning to infer private information when acting
 - ❑ Signaling intertwines strategies
 - ❑ Nonclassical information structures make sequential decomposition (DP) difficult
 - ❑ Characterization of *Information State* allows for sequential decomposition
- ❑ Concurrent policy adaptation => non-stationary environment for the agents
- ❑ Teams simpler than games: No (strategic) deviations to be analyzed
- ❑ High solution complexity: NEXP complete, see Witsenhausen counterexample too
 - ❑ Few benchmarks, but using *D. Tang et al. 2023* more examples can be solved

Solutions for cooperative multi-agent problems

- ❑ (Conceptual) Approaches exist to solve many multi-agent team problems
 - ❑ Person-by-person approach (*Witsenhausen1979*)
 - ❑ Designer's approach (*Witsenhausen1973*)
 - ❑ Common-Information (CI) approach (*NayyarMahajanTenketzis2014*)
- ❑ CI Approach
 - ❑ Transforms problem to a (fictitious) coordinator's POMDP using CI
 - ❑ Coordinator uses CI to provide prescriptions to each agent
 - ❑ Prescription: Partial function from private history to actions
 - ❑ Prescriptions are then actions of coordinator
 - ❑ Equivalence to team problem holds with deterministically chosen prescriptions
 - ❑ Each agent instantiates the coordinator and solves the problem
- ❑ Provides Strategy Independent but Model Dependent *Information State*!

Exemplary Problem*

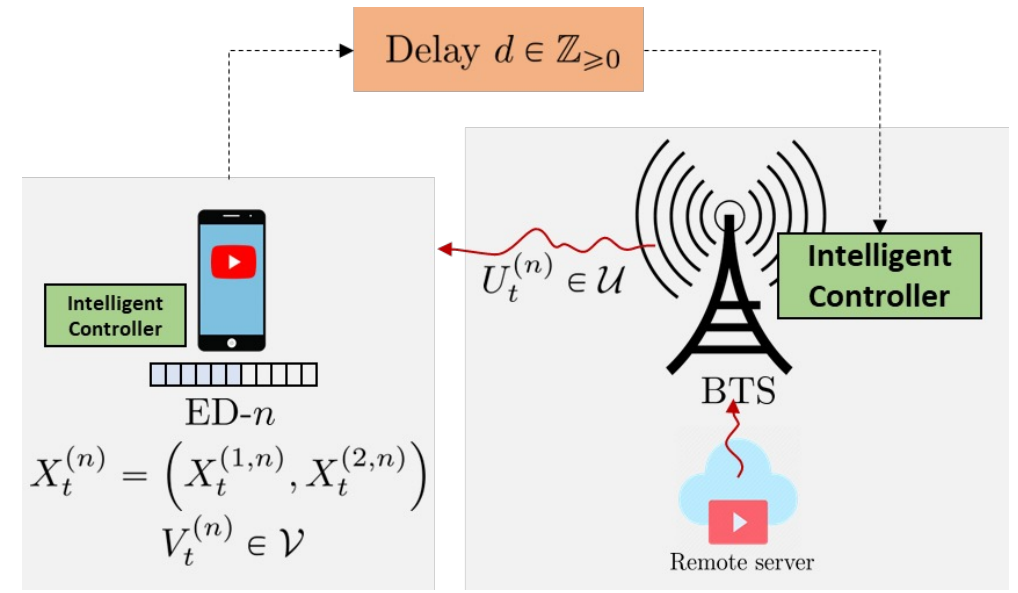
- ❑ Controller at a Base-Transceiver-Station (BTS) sends video-packets to Edge-Devices (EDs)
- ❑ **Transmission-energy constraint** at the BTS
- ❑ EDs play video packets and send application-level feedback to the BTS
- ❑ **Application-level feedback** is available at the BTS controller after some information-processing and/or **feedback-channel delay**
- ❑ A team problem with the (common) goal of providing a high quality of video streaming experience to end-users



*N. Khan, et al. IEEE INFOCOM 2024

System Model I

- N EDs and 1 BTS
- Discrete time setting ($t = 1, 2, \dots$)
- State of ED- n at time t , $X_t^{(n)} = (X_t^{(1,n)}, X_t^{(2,n)})$
 - $X_t^{(n)} \in \mathcal{X}$, finite
 - $X_t^{(1,n)}$: Application-state of ED- n at time t
 - $X_t^{(2,n)}$: QoE tracking metric for ED- n at time t
 - States of all EDs, $X_t \triangleq X_t^{(1:N)}$
- Play-out action of ED- n at time, $V_t^{(n)}$
 - $V_t^{(n)} \in \mathcal{V}$, finite
 - Play-out actions of all EDs, $V_t \triangleq V_t^{(1:N)}$
- Transmission actions of BTS, $U_t = U_t^{(1:N)}$
 - $U_t^{(n)} \in \mathcal{U}$, finite
- **No common information between agents**



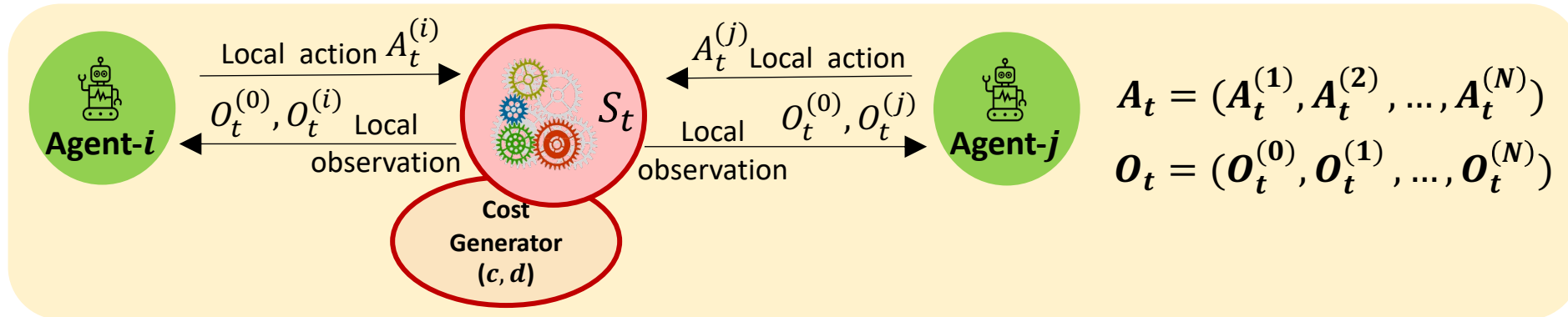
- Client-Feedback
 - Each ED communicates its state and action to the BTS which reaches it with a fixed deterministic delay $d \in \mathbb{Z}_{\geq 0}$
 - **Feedback of ED- n is known to it and the BTS only, and not to the rest of the EDs**

Multi-agent Constrained Dec POMDP

Decision Process

- ❑ N number of cooperative agents who take actions sequentially in a discrete time setting ($t = 1, 2, \dots$)
- ❑ Environment with an internal state $\mathbf{S}_t$ that lives in a state space $\mathcal{S}$
- ❑ After all agents have formed a joint-action $\mathbf{A}_t = \mathbf{A}_t^{(1:N)}$ (based on their own information)
 - ❑ State of the environment evolves to $\mathbf{S}_{t+1}$
 - ❑ Agents receive parts of joint-observation $\mathbf{O}_{t+1} = \mathbf{O}_t^{(0:N)}$, and

$O_t^{(0)}$ is observed by all agents.
 $O_t^{(n)}$ is observed by agent- n only.
 - ❑ Team incurs immediate costs (*in a countable space for learning context*):
 - ❑ Immediate objective cost $c(\mathbf{S}_t, \mathbf{A}_t) \in \mathbb{R}$ (for the primary objective)
 - ❑ Immediate constraint cost $d(\mathbf{S}_t, \mathbf{A}_t) \in \mathbb{R}^K$ (for all other constraint-related objectives)



Decision Process

- State space $\mathcal{S}$ is a general topological space
- Factorized and countable Joint-Observation space, $\mathcal{O} = \prod_{n=0}^N \mathcal{O}^{(n)}$
- Factorized and finite Joint-Action space, $\mathcal{A} = \prod_{n=1}^N \mathcal{A}^{(n)}$

Transition law: $\mathcal{P}_{tr} \triangleq \{P_{saBo}\}$

$$P_{saBo} = \mathbb{P}(S_{t+1} \in B, O_{t+1} = o | S_t = s, A_t = a).$$

Next state

Borel-measurable
set in $\mathcal{S}$

Current state

Next
joint-observation

Current
joint-action

Initial Distribution:

$$P_1 \in M_1(\mathcal{S} \times \mathcal{O}).$$

Set of all probability measures on
Borel σ -algebra of $\mathcal{S} \times \mathcal{O}$ endowed
with the topology of weak-convergence.

Optimization Problem

Policy-Profile Space

Common History: $H_t^{(0)} \triangleq (O_{1:t}^{(0)})$.

Private History of agent- n :

$$H_1^{(n)} \triangleq O_1^{(n)} \setminus O_1^{(0)}, \text{ and}$$

$$H_t^{(n)} \triangleq (H_{t-1}^{(n)}, A_{t-1}^{(n)}, O_t^{(n)}) \setminus H_t^{(0)} \quad \forall t \geq 2.$$

Policy space of agent- n : $\mathcal{U}^{(n)}$.

- Each member of $\mathcal{U}^{(n)}$ is of the form $u_{1:\infty}^{(n)}$.
- Agent- n uses the distribution $u_t^{(n)}(H_t^{(0)}, H_t^{(n)})$ to choose its action $A_t^{(n)}$.

Policy-Profile space: $\mathcal{U} = \prod_{n=1}^N \mathcal{U}^{(n)}$.

- u denotes a typical member of $\mathcal{U}$.

Objective and Constraint Functions

Expected Infinite-Horizon Discounted Objective Cost:

$$C(u) \triangleq \mathbb{E}_{P_1}^{(u)} \left[\sum_{t=1}^{\infty} \alpha^{t-1} c(S_t, A_t) \right].$$

Expected Infinite-Horizon Discounted Constraint Cost:

$$D(u) \triangleq \mathbb{E}_{P_1}^{(u)} \left[\sum_{t=1}^{\infty} \alpha^{t-1} d(S_t, A_t) \right].$$

Mixtures: $\mathcal{M}_1(\mathcal{U})$. Set of all probability measures on Borel σ -algebra of $\mathcal{U}$.

- μ denotes a typical member of $\mathcal{M}_1(\mathcal{U})$.

$\mathcal{M}_1(\mathcal{U})$ to be described by giving $\mathcal{U}$ a suitable topology.

- Can define extended versions of long-term costs.

$$\hat{C}(\mu) \triangleq \mathbb{E}^{(U \sim \mu)} [C(U)].$$

$$\hat{D}(\mu) \triangleq \mathbb{E}^{(U \sim \mu)} [D(U)].$$

Optimization Problem

Optimization Problem

minimize $\hat{C}(\mu)$
subject to $\mu \in \mathcal{M}_1(\mathcal{U})$ and $\hat{D}(\mu) \leq \check{D}$. (MA-C-POMDP)

$\underline{C}$ = optimal value of (MA-C-POMDP).
Set $\underline{C}$ to ∞ if no feasible mixture exists.

Lagrangian function: Let $\mathcal{Y} \triangleq \{\lambda \in \mathbb{R}^K : \lambda \geq 0\}$. Define

$$\begin{aligned} \hat{L} : M_1(\mathcal{U}) \times \mathcal{Y} &\rightarrow \mathbb{R} \cup \{\infty\} \\ (\mu, \lambda) &\mapsto \hat{C}(\mu) + \langle \lambda, \hat{D}(\mu) - \check{D} \rangle \end{aligned}$$

Is the Lagrangian formulation useful?

Optimization Problem: Comments

Optimization Problem

minimize $\hat{C}(\mu)$
subject to $\mu \in \mathcal{M}_1(\mathcal{U})$ and $\hat{D}(\mu) \leq \check{D}$. (MA-C-POMDP)

$\underline{C}$ = optimal value of (MA-C-POMDP).
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Lagrangian function: Let $\mathcal{Y} \triangleq \{\lambda \in \mathbb{R}^K : \lambda \geq 0\}$. Define

$$\begin{aligned} \hat{L} : M_1(\mathcal{U}) \times \mathcal{Y} &\rightarrow \mathbb{R} \cup \{\infty\} \\ (\mu, \lambda) &\mapsto \hat{C}(\mu) + \langle \lambda, \hat{D}(\mu) - \check{D} \rangle \end{aligned}$$

❑ Do not have an LP on occupation-measures (*Altman1999, Borkar2002*)

❑ The general decentralized setting considered does not yield a characterization of occupation-measures on the state and joint-action, as in MDPs

❑ CI approach needs technical innovations

❑ Leads to uncountable state-space in the equivalent belief-based constrained MDP

❑ Constrained MDPs and POMDPs results need countable state-space (*Altman1999, Borkar2002*)

Key Assumptions

□ Assumption 1:

Immediate objective cost $\mathbf{c}$ is bounded from below

□ Assumption 2:

Immediate constraint costs $\mathbf{d}$ are bounded both from above and below

□ Assumption 3 (Slater's condition):

There exists a mixture (policy) $\bar{\mu} \in M_1(\mathbf{u})$ and $\zeta > 0$ for which

$$\hat{D}(\bar{\mu}) \leq \check{D} - \zeta \mathbf{1}$$

Strong Duality Result

Result 1: Strong Duality

Under Assumptions 1 and 2, **strong duality** holds i.e.,

$$\underline{C} = \inf_{\mu \in M_1(\mathcal{U})} \sup_{\lambda \in \mathcal{Y}} \hat{L}(\mu, \lambda) = \sup_{\lambda \in \mathcal{Y}} \inf_{\mu \in M_1(\mathcal{U})} \hat{L}(\mu, \lambda).$$

Moreover, there exists a $\mu^* \in M_1(\mathcal{U})$ such that

$$\underline{C} = \sup_{\lambda \in \mathcal{Y}} \hat{L}(\mu^*, \lambda).$$

Result on Existence of Saddle-Point

Result 2: Existence of Saddle-Point

Under Assumptions 1, 2, and 3, result 1 (strong duality) holds and there exist $\mu^* \in M_1(\mathcal{U})$ and $\lambda^* \in \mathcal{Y}$ such that for all $(\mu, \lambda) \in M_1(\mathcal{U}) \times \mathcal{Y}$,

$$\hat{L}(\mu^*, \lambda) \leq \hat{L}(\mu^*, \lambda^*) = \underline{C} \leq \hat{L}(\mu, \lambda^*),$$

i.e., μ^* minimizes $\hat{L}(\cdot, \lambda^*)$ and λ^* maximizes $\hat{L}(\mu^*, \cdot)$.

In addition, the primal dual pair (μ^*, λ^*) satisfies **complementary-slackness**:

$$\langle \lambda^*, D(\mu^*) - \check{D} \rangle = 0.$$

Summary of General Results and Core Questions

- ❑ Studied **MA-C-POMDPs** for infinite-horizon discounted costs
- ❑ Showed results on **strong duality** and **existence of saddle point**
 - ❑ The results hold over policy-profile mixture space, $M_1(\mathcal{U})$
 - ❑ Joint mixing of policies! Q: Is this needed? YES! (Altman book example can be used.)
 - ❑ Mixing is carried out at time 0 (before interacting with the environment)
 - ❑ Mixing is needed to ensure tightness of joint constraints
 - ❑ Question: How about using a behavioral policy?
 - ❑ Question: Any characterization of the number of randomizations?
 - ❑ Question: How to use learning-based schemes?
- ❑ Complex optimization problem: No occupancy measures characterization

Further Structural Insights

High Level Overview of Agent Interactions

- Agents use **interaction-mechanisms** to interact with the environment.
- We are already familiar with several **interaction-mechanisms in single-agent fully-observed systems**.

$N = 1$

Behavioral policies	$\mathcal{U}^{(1)}$	$u_{1:\infty}^{(1)}$	$u_t^{(1)} : \mathcal{H}_t^{(0)} \times \mathcal{H}_t^{(1)} \rightarrow M_1(\mathcal{A}^{(1)}).$
Pure policies	$\mathcal{U}_{\text{pure}}^{(1)} \subseteq \mathcal{U}^{(1)}$	$u_{1:\infty}^{(1)}$	$u_t^{(1)} : \mathcal{H}_t^{(0)} \times \mathcal{H}_t^{(1)} \rightarrow \mathcal{A}^{(1)}.$
Markov policies	$\mathcal{U}_{\text{Markov}}^{(1)} \subseteq \mathcal{U}^{(1)}$	$u_{1:\infty}^{(1)}$	$u_t^{(1)}(h_t^{(0)}, h_t^{(1)}) = u_t^{(1)}(s_t).$
Stationary policies	$\mathcal{U}_{\text{stat}}^{(1)} \subseteq \mathcal{U}_{\text{Markov}}^{(1)}$	$u_{1:\infty}^{(1)}$	$u_t^{(1)} = u_1^{(1)} \text{ for all } t.$

- Can consider several more.

Mixed Behavioral policies	$M_1(\mathcal{U}^{(1)}).$
Mixed Pure policies	$M_1(\mathcal{U}_{\text{pure}}^{(1)}).$
$\vdots$	$\vdots$

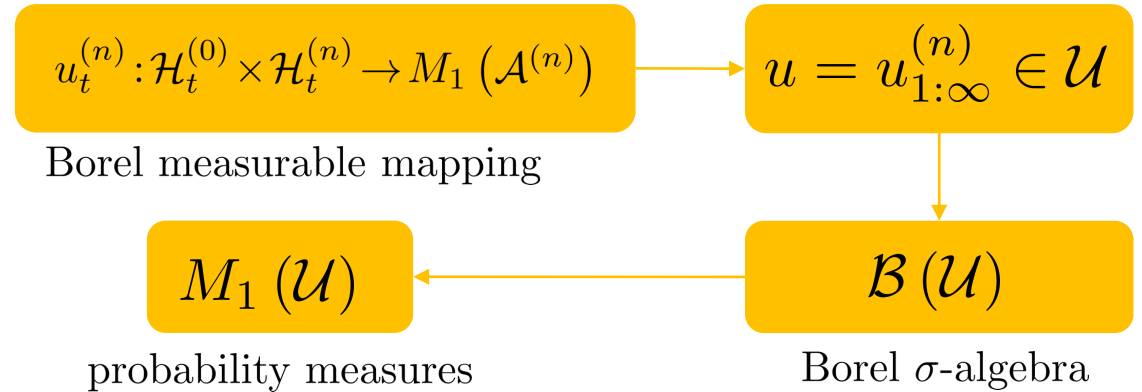
With imperfect recall but no absentmindedness.

With imperfect recall and absentmindedness.

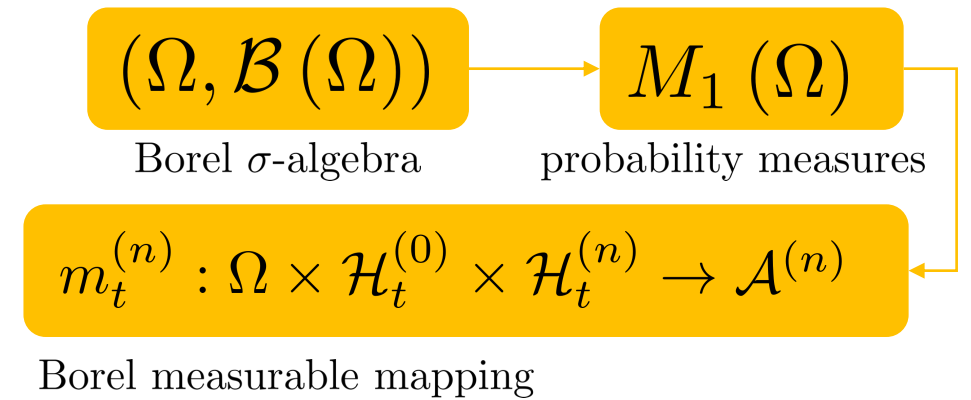
Core Aspects of Interaction-Mechanisms

■ Mixing: Randomization Methods

- From decision-rules to randomization devices / probability distributions.
 - Prevalent in stochastic control.



- From randomization devices / probability distributions to decision-rules.
 - Prevalent in game theory.



Core Aspects of Interaction-Mechanisms

■ Memory: Domains of Decision-rules

$$u_t^{(n)} : \mathcal{H}_t^{(0)} \times \mathcal{H}_t^{(n)} \rightarrow M_1(\mathcal{A}^{(n)})$$

Perfect recall: The decision-rules are based on agent's entire action-observation history [5].

Imperfect recall: Agent forgets her action-observation history.

Without Absentmindedness [7,8]

An agent can choose a distinct decision rule for each time-step.

With absentmindedness

An agent must choose a decision-rule for more than one time-step.

$$u_t^{(n)} : \theta_t^{(n)} \left(\mathcal{H}_t^{(0)} \times \mathcal{H}_t^{(n)} \right) \rightarrow M_1(\mathcal{A}^{(n)})$$

Interaction-mechanisms based on (approximate) information states [6] are examples of imperfect recall.

$$u_t^{(n)} = u_{t'}^{(n)} \text{ for some } t \neq t'.$$

High Level Overview of Agent Interactions

- Agents use **interaction-mechanisms** to interact with the environment.
- For **multi-agent partially-observed systems**. (Each agent's full history needs to be used OR compressed carefully).

Decentralized Behavioral policy-profiles

Decentralized Pure policy-profiles

Individually Mixed Behavioral policy-profiles

Individually Mixed Pure policy-profiles

(Jointly) Mixed Behavioral policy-profiles

(Jointly) Mixed Pure policy-profiles

Behavioral Coordination Policies

Pure Coordination Policies

Mixed Behavioral Coordination Policies

Mixed Pure Coordination Policies

MA-AIS based Behavioral Coordination Policies

MA-AIS based Pure Coordination Policies

Mixed MA-AIS based Behavioral Coordination Policies

Mixed MA-AIS based Pure Coordination Policies

MA-AIS based stationary policies

$$\mathcal{U} = \prod_{n=1}^N \mathcal{U}^{(n)}$$

$$\mathcal{U}_{\text{pure}} = \prod_{n=1}^N \mathcal{U}_{\text{pure}}^{(n)}$$

$$\prod_{n=1}^N M_1(\mathcal{U}^{(n)})$$

$$\prod_{n=1}^N M_1(\mathcal{U}_{\text{pure}}^{(n)})$$

$$M_1(\mathcal{U})$$

$$M_1(\mathcal{U}_{\text{pure}})$$

$$\mathcal{V}$$

$$\mathcal{V}_{\text{pure}}$$

$$M_1(\mathcal{V})$$

$$M_1(\mathcal{V}_{\text{pure}})$$

$$\mathcal{W}$$

$$\mathcal{W}_{\text{pure}}$$

$$M_1(\mathcal{W})$$

$$M_1(\mathcal{W}_{\text{pure}})$$

$$\mathcal{W}_{\text{stat}}$$

With perfect recall

With imperfect recall but no
absentmindedness.

With imperfect recall and absentmindedness.

A Coordination Perspective on Mixed Policy-Profiles

$$M_1(\mathcal{U}) \equiv M_1(\mathcal{U}_{\text{pure}}) \equiv M_1(\mathcal{V}_{\text{pure}}) \equiv M_1(\mathcal{V}) \equiv \mathcal{V}.$$

Mixed (behavioral/pure) policy-profiles

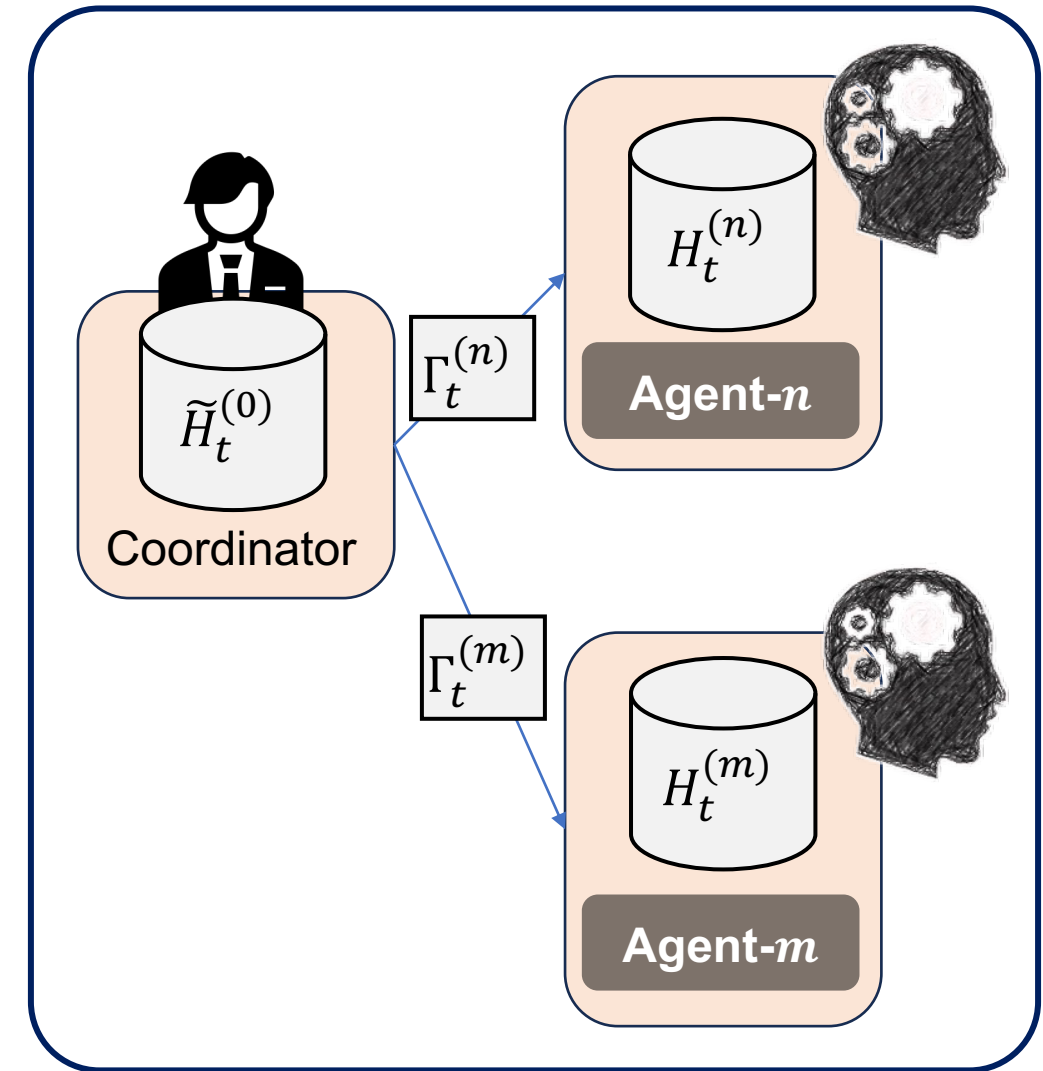
Mixed (behavioral/pure)
coordination policies

Behavioral
coordination
policies



Coordination Policies and Coordinated System

- Each agent instantiates a virtual agent, known as *coordinator* [9,10].
 - Information of each copy of the coordinator is ensured to be same throughout the team.
- Coordinator gets to see the common observations, $O_t^{(0)}$ s, but does not get to see the agents' private observations and actions, $O_t^{(1:N)}$ s and $A_t^{(1:N)}$ s.
- Agents are passive; they follow the *prescriptions* of the coordinator.



Coordination Policies and Coordinated System

Coordinator's Prescriptions:

- A tuple of N mappings given by $\mathbf{\Gamma}_t = \mathbf{\Gamma}_t^{(1:N)}$, where $\Gamma_t^{(n)}$ maps $\mathcal{H}_t^{(n)}$ to $\mathcal{A}^{(n)}$.

- $\Gamma_t^{(n)}$ is an enforcing instruction for agent n . Agent n chooses action according to the rule,

$$A_t^{(n)} = \Gamma_t^{(n)} \left(H_t^{(n)} \right).$$

- Coordinator faces a **SA-C-POMDP**.
 - Can define **behavioral policies for the coordinator** based on her **prescription-observation histories**.

$\Gamma_t^{(n)}$	
Private History	Prescribed Action
$\vdots$	$\vdots$
$h_t^{(n)}$	$a_t^{(n)}$
$\vdots$	$\vdots$

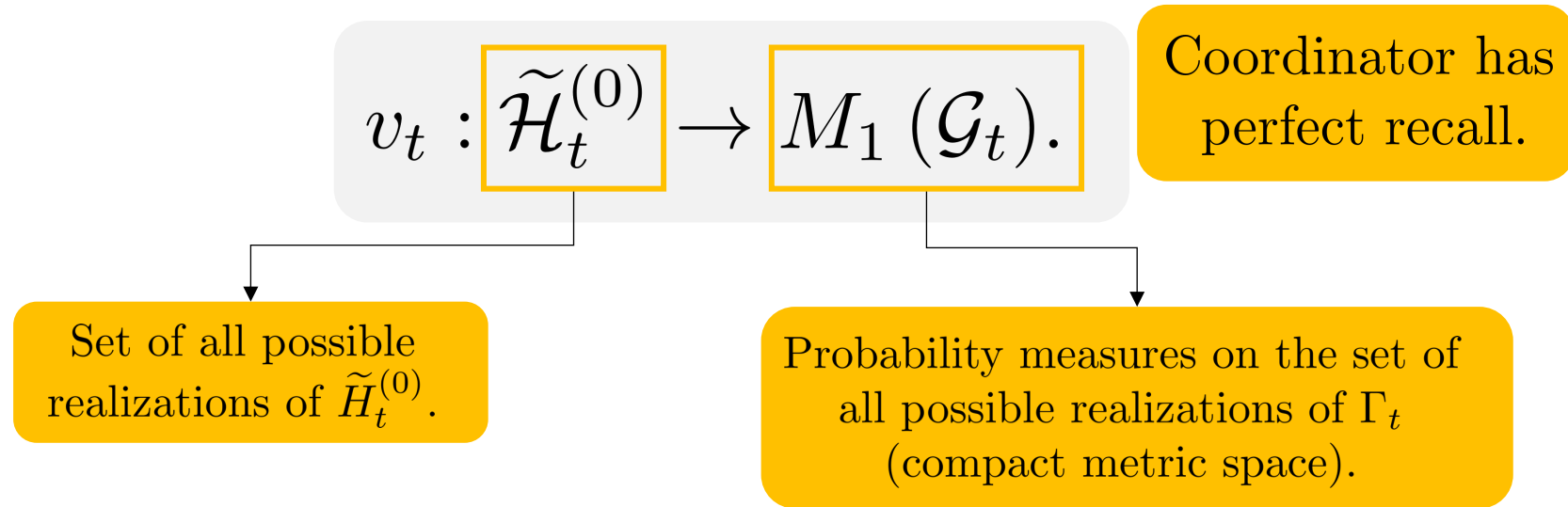
Prescription-Observation Histories

$$\tilde{H}_1^{(0)} := O_1^{(0)} \text{ and}$$

$$\tilde{H}_t^{(0)} := \left(\tilde{H}_{t-1}^{(0)}, \Gamma_{t-1}, O_t^{(0)} \right) \text{ for all } t \in [2, \infty]_{\mathbb{Z}}.$$

Coordination Policies and Coordinated System

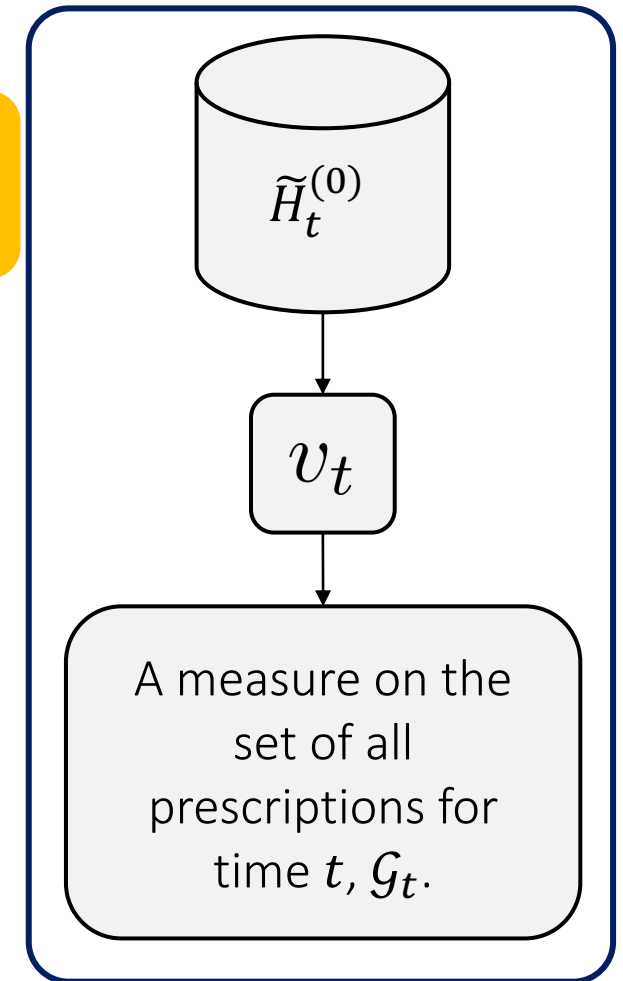
Coordinator's Decision-rules



Coordinator's Behavioral Policy:

$$v = v_{1:\infty} = (v_1, v_2, \dots).$$

$\mathcal{V}$



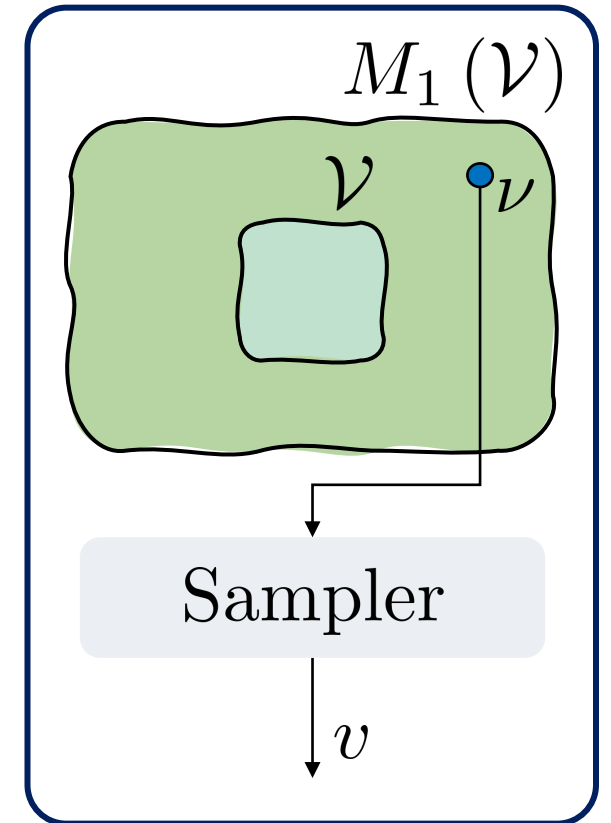
Coordination Policies and Coordinated System

■ Coordination Policies for the Team:

- $\mathcal{V}$: Behavioral coordination policies.
- $\mathcal{V}_{\text{pure}}$: Pure coordination policies.
- $M_1(\mathcal{V})$: Mixed behavioral coordination policies.
- $M_1(\mathcal{V}_{\text{pure}})$: Mixed pure coordination policies.

Optimization Problem

$$\begin{aligned}
 &\text{minimize } C_T^{(\alpha)}(\nu) \\
 &\text{s.t. } \nu \in M_1(\mathcal{V}), \text{ and } D_T^{(\alpha)}(\nu) \leq \check{D}. \quad (COP-2)
 \end{aligned}$$



We show that: *COP-2* is the same as *COP-1*.

Results on Coordination Policies

Definition 3.1: Dominance and Equivalence

- Consider two sets of interaction-mechanisms W and W' .
- W dominates W' for time-horizon T , i.e., $W \succeq_T W'$, if
 $\forall w' \in W', \exists w = w(w') \in W$ such that $\forall \alpha, C_T^{(\alpha)}(w) \leq C_T^{(\alpha)}(w')$ and $D_T^{(\alpha)}(w) \leq D_T^{(\alpha)}(w')$.
- W and W' are equivalent for time-horizon T , i.e., $W \equiv_T W'$, if $W \succeq_T W'$ AND $W' \succeq_T W$.
- If W dominates W' for all time-horizons, we write $W \succeq W'$.
- If W and W' are equivalent for all time-horizons, we write $W \equiv W'$.

We will show $M_1(\mathcal{U}) \equiv M_1(\mathcal{U}_{\text{pure}}) \equiv M_1(\mathcal{V}_{\text{pure}}) \equiv M_1(\mathcal{V}) \equiv \mathcal{V}$.

Mixed (behavioral/pure) policy-profiles

Mixed (behavioral/pure)
coordination policies

Behavioral
coordination
policies

Results on Coordination Policies

Theorem 3.1: Relations between Interaction Mechanisms

Assume

- “bounded-from-below immediate costs”, AND
- “finite number of common and private observations.”

Altman book example shows

$$M_1(\mathcal{U}) \not\equiv \mathcal{U}.$$

Then, the following equivalences hold.

$$1. \mathcal{U} \equiv \prod_{n=1}^N M_1(\mathcal{U}_{\text{pure}}^{(n)}) \equiv \prod_{n=1}^N M_1(\mathcal{U}^{(n)}),$$

Mixed and behavioral policy-profiles

$$2. M_1(\mathcal{U}) \equiv M_1(\mathcal{U}_{\text{pure}}),$$

$$3. M_1(\mathcal{U}_{\text{pure}}) \equiv M_1(\mathcal{V}_{\text{pure}}), \text{ and}$$

Mixed policy-profiles and mixed coordination policies

$$4. M_1(\mathcal{V}_{\text{pure}}) \equiv M_1(\mathcal{V}) \equiv \mathcal{V}.$$

Mixed and behavioral coordination policies

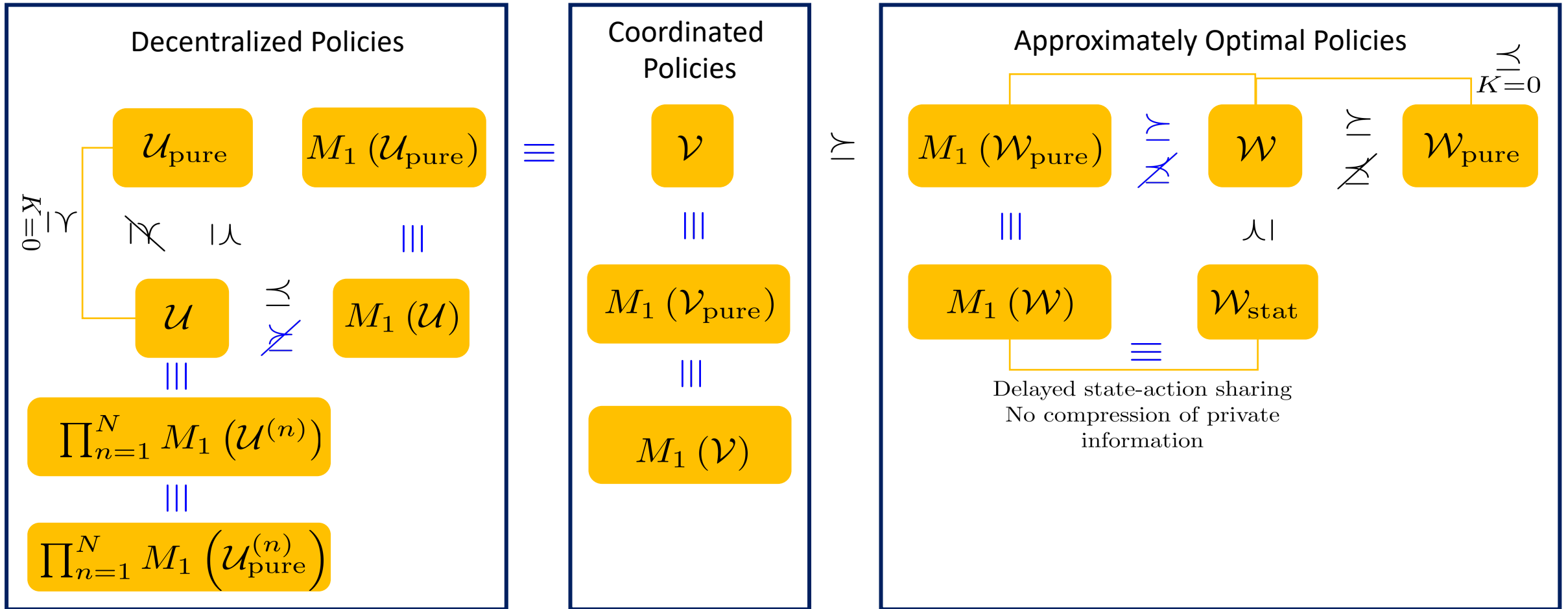
Thus, in particular, $M_1(\mathcal{U}) \equiv \mathcal{V}$.

Overview of other results

- ❑ Coordinator viewpoint
 - ❑ Yields a single-agent constrained MDP on a countable state space
 - ❑ Can apply Borkar2002's results on number of randomizations
 - ❑ Can develop primal-dual method for solution – Impractical!
 - ❑ Needs model knowledge and complexity grows doubly exponentially in time
- ❑ Multi-agent Approximate Information State
 - ❑ Coordinator's viewpoint allows for generalizing Hsu Kao & S. [AISTATS 2022] to constrained problems
 - ❑ Approximations use imperfect recall
 - ❑ Mixed strategies outperform behavioral strategies
 - ❑ Provable performance guarantees developed + smoothness in approximations

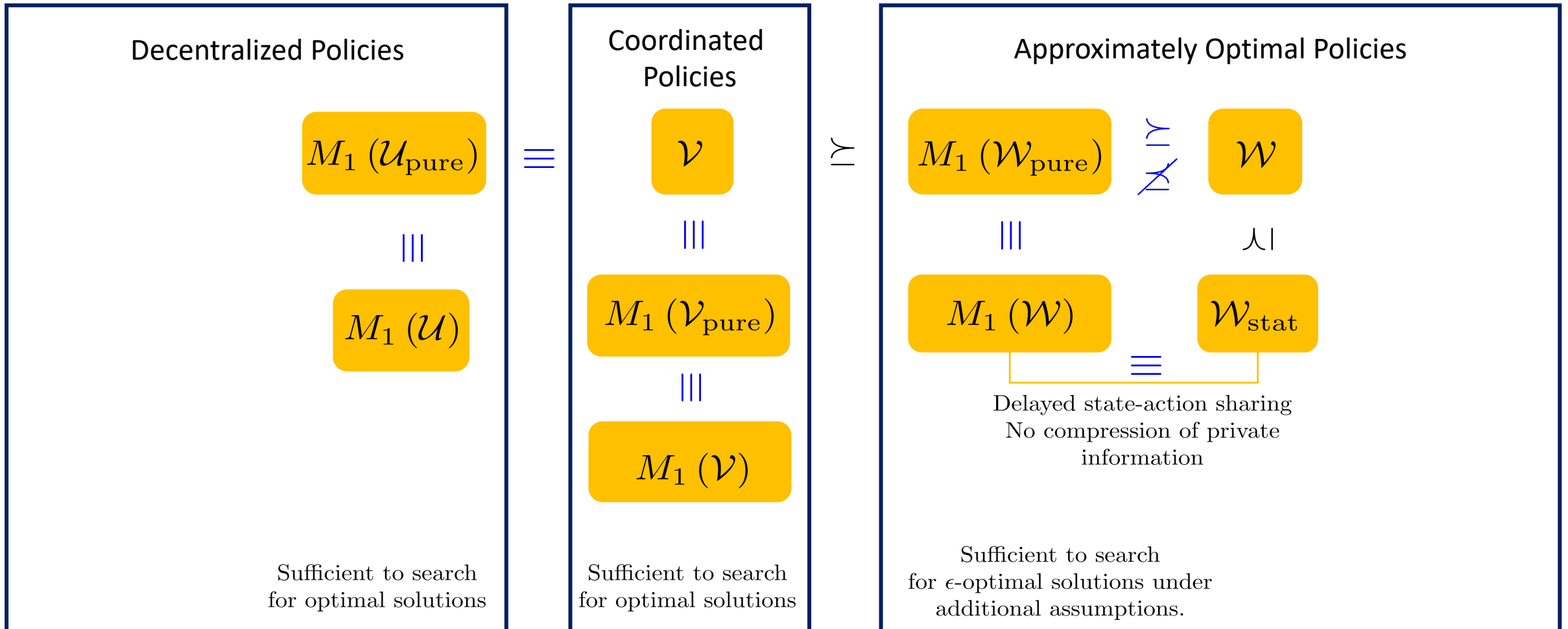
Summary and Key Takeaways

Dominance and Equivalence Relations



Summary and Key Takeaways

Sufficiency for Search of ϵ -Optimal Solutions in constrained settings ($K > 0$).



Extensions to dynamic games?

- ❑ *Information State* hard to characterize in general
 - ❑ *D. Tang et al. DGAA 2023, D. Tang PhD thesis 2021, & D. Tang et al. DGAA 2025*
- ❑ For some models, [Model](#) and [Strategy Independent](#) *Information States* exist
 - ❑ Equilibrium using strategies with compressed information exists
 - ❑ All equilibrium payoff profiles can be preserved
 - ❑ In general, the domain grows with time!
- ❑ (CI-based) Belief State as candidate *Information State* as in teams?
 - ❑ In many cases, this Belief State has a time-invariant domain
 - ❑ In general, this Belief State is [Model](#) and [Strategy Dependent](#)
 - ❑ *Equilibria may not exist in general, and sequential decomposition may not hold*

Thank You

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